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Investigating explainability of deep learning models for sequential data on stock price prediction

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Abstract

Accurate stock price prediction remains a significant challenge within the financial sector, where deep learning models have shown considerable predictive power but suffer from limited interpretability due to their inherent "black box" nature. This study addresses this issue by integrating Explainable Artificial Intelligence (XAI) techniques with Conv1D and LSTM deep learning models to enhance transparency in stock price forecasting. We propose a novel XAI algorithm tailored for deep learning, utilizing post-hoc explanation methods such as LIME and SHAP to elucidate the decision-making processes of the models. In addition, we evaluate the predictive performance of Conv1D and LSTM models against state-of-the-art techniques, including NBEATS and LTCN, revealing that NBEATS consistently achieves superior performance across multiple datasets. Our findings highlight the predictive significance of stock price of Adani power equity from the previous two days of stock price, whereas data from two days prior shows reduced influence on predictive accuracy. The study demonstrates that XAI-augmented Conv1D and LSTM models strike a balance between transparency and predictive performance, offering investors more interpretable insights into market trends. This research underscores the importance of explainable stock price prediction and calls for continued exploration of this emerging domain.

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1. Introduction

One technology that has completely changed the world with its performance and potential is artificial intelligence. The industrial and internet revolutions profoundly altered our perceptions of the world, and the intelligence revolution is about to occur in the modern world. Artificial intelligence is briskly extending throughout several industries with exceptional efficacy in addressing complex real-world challenges. From rudimentary image classification to sophisticated DNA sequence analysis, AI-driven methodologies have demonstrated significant potential. However, the inherent opacity of contemporary AI models, particularly deep learning architectures, poses a substantial limitation. The term "black box" has been applied to models whose internal decision-making processes remain opaque to human comprehension, thereby limiting the ability to ascertain the logic underpinning their outputs. Explainable AI solutions such as fuzzy inference systems are utilized to tackle the problem of opaque AI systems by virtue of their interpretability[1][2][3][4][5]

Artificial intelligence is also being used by the banking industry to increase revenues by optimizing the performance of these models. But the stock market is still quite complicated today. Market prediction becomes an enormous undertaking when it is influenced by n-dimensional elements, which might range from investor emotion to world events. Traditional models struggle to understand complex relationships because of this intricacy. However, in terms of performance and outcomes, more recent methods such as deep learning and machine learning models have surpassed older models. However, these models are not interpretable because they are black box models. Concerns concerning AI models' dependability and usefulness are raised by their lack of transparency.[6][1]

While traditional forecasting models like statistical models Arima and Sarima can be interpreted, they are not so effective and scalable in identifying complex relationships. In contrast, deep learning models like RNN and LSTM are effective at identifying intricate patterns and relationships, but they cannot be interpreted. Consequently, our goal is to close this gap by employing interpretable stock price forecast.[7][8][9][10][11]

Interpretable time series prediction bridges this gap. It aims to develop models that not only generate forecasts but also provide insights into the reasoning behind them. Imagine a model predicting a surge in patient admissions. An interpretable model would not just predict the surge, but also pinpoint factors like a flu outbreak or upcoming holiday that contribute to it. This empowers users to build trust by understanding the rationale behind predictions, users can have more confidence in the model's output. Better forecasts can result from debugging and improving insights into significant aspects that may assist detect potential biases or inaccuracies in the model. Take informed actions about Knowing the root causes behind predictions allows for proactive decision-making. Which have high significance in the age of online social networks [12][13][14]

The document is organized as follows: In Section II, various methodologies for Artificial Intelligence (AI) and Explainable AI (XAI) in time series prediction are reviewed. Section III presents the proposed algorithm along with a detailed description of the dataset and the methodology employed. Section IV focuses on the evaluation of the models and the implementation of XAI techniques. Finally, Section V concludes the document by summarizing key findings and outlining potential directions for future research

2. Literature Review

2.1. AI in time series prediction

Recent studies have developed an effective stock price forecast Time-series Recurrent Neural Network (TRNN). The proposed approach processes the time series data using sliding windows and establishes trade volume. Data compression is achieved, and financial market-related factors are used to determine trends and turning moments. An augmented two-dimensional RNN model demonstrates a heightened correlation between contemporaneous stock valuation and preceding trade volume relative to a unidimensional framework.. The TRNN model is established by processing and compressing the trade volume data, ensuring accuracy and efficiency.[15]

Building upon the foundation of recurrent neural networks, this research investigates the efficacy of long short-term memory (LSTM) architectures in capturing temporal dependencies for time series forecasting. Their study made use

of a wide range of real-world data and offered quantitative insights into LSTM performance. Efficiency in a variety of time-series forecasting activities [8]

Recent advancements in deep learning have enabled significant improvements in time series prediction tasks, particularly through the application of hybrid models combining various architectures. Studies have explored combinations of convolutional neural networks (CNNs), long short-term memory networks (LSTMs), and transformer (TF) models, aimed at leveraging the strengths of each architecture for more accurate predictions. For instance, one line of research systematically experimented with all possible combinations of CNN, LSTM, and transformer models, comparing the hybrid architectures to single-model approaches across a range of optimizers.[16]

The increasing complexity of financial time series data has driven the development of innovative deep learning models that merge different methodologies to capture the intricacies of such data more effectively. One promising approach is the Temporal Convolution and Linear Model (TCLM), which addresses the dual challenges of handling irregularities and modeling linear trends inherent in financial time series. By combining temporal convolutional layers, which excel at extracting features from sequential data, with a linear model that accurately tracks long-term trends, TCLM provides a more holistic framework for time series analysis. This hybrid model represents a significant advancement in the quest for more accurate and reliable financial forecasting.[17]

The field of deep learning continues to reshape the landscape of time series forecasting, where innovative architectures like N-BEATS have emerged as top-performing solutions for univariate prediction tasks. One promising enhancement to the N-BEATS framework tackles the point forecasting challenge by introducing parametric probabilistic forecasts. This modification enriches the predictive capability by incorporating uncertainty estimates, making the models particularly valuable in practical applications where quantifying risk is essential.[18]

A growing body of work in deep learning for time series has focused on designing models that effectively capture spatial-temporal relationships within complex datasets. One notable contribution is the attention-based convolutional closed-form continuous-time neural network (AC-CfC), which has been proposed for emotion recognition based on raw EEG data. This model introduces a channel attention mechanism that assigns dynamic weights to the EEG channels, capturing the intricate dependencies between them, which is crucial for accurate spatial modeling of EEG signals. The AC-CfC then utilizes one-dimensional convolutional neural networks and closed-form continuous-time neural networks to extract deep spatial-temporal features, allowing it to effectively handle the temporal dependencies inherent in EEG data. To further improve classification performance, particularly for classes that are difficult to distinguish, the model incorporates an adaptive loss-controlling mechanism. This mechanism dynamically adjusts the loss function, enhancing the model's ability to make accurate decisions between easily confused classes.[19]

2.2. Explainable AI in time series prediction

While predictive accuracy is crucial, understanding the underlying reasons for model decisions is equally important. To anticipate interval-valued stock values, an innovative interval dual convolutional neural network (Dual-CNN) model-based technique was put forth. Individual and pertinent stock data are gathered, then converted into pictures. Next, a model called Dual CNN is put out to forecast interval valued stock values. In particular, two distinct convolutional neural network (CNN) models are built to extract individual stock characteristics and pertinent stock features, respectively. For the purpose of making an interval valued stock price forecast, an interval multilayer perceptron model is subsequently employed. [20]

To enhance the interpretability of time series models, Three distinct machine learning models, a nonlinear ensemble technique, and a decomposition algorithm comprised a unique two-stage prediction model that was proposed. By using variational mode decomposition (VMD), The stock price time series is disaggregated into a restricted set of sub-series. Subsequently, SVR, ELM, and DNN models are individually applied to forecast the decomposed sub-series. Predictions from individual models on sub-series are aggregated to form the initial stock price forecast. A nonlinear ensemble technique based on expectation-maximization (ELM) is used to aggregate initial stock price forecasts.[21]

Expanding the scope beyond financial markets, To forecast trends in carbon prices, a machine learning model that is interpretable is suggested. The methodology incorporates a five-pronged approach consisting of two-stage feature selection, LightGBM enhanced by Bayesian optimization, SHAP, and CEEMDAN. The carbon prices from the Hubei carbon trading market, the biggest market in China, are used to verify the efficacy of the suggested model.[22]

The application of explainable AI (XAI) has gained traction in various financial domains. Recent research introduced a unique strategy that makes use of Explainable AI (XAI) and Federated Learning (FL). By cooperating to

jointly train a model to detect fraudulent transactions, financial institutions may preserve data privacy and confidentiality without directly exchanging customer data. Concurrently, the incorporation of XAI guarantees that the model's forecasts may be comprehended and evaluated by human specialists, hence augmenting the system's transparency and reliability. [23]

In the context of blockchain transactions, due to the rarity of unlawful transactions, anomaly identification in blockchain transaction data is a difficult operation. Despite a number of research in the area, one problem still exists: there are no justifications for the predictions made by the model. The research addressed shortcomings by incorporating anomaly rules and eXplainable Artificial Intelligence (XAI) approaches into ensemble classifiers that are based on trees and are used to identify unusual Bitcoin transactions. The contribution of each feature is measured using the Shapley Additive exPlanation (SHAP) approach, which works well with ensemble models. [24]

Moving beyond fraud detection and blockchain transactions, A hybrid approach combining machine learning and natural language processing was suggested to assess the authenticity of social media finance content creators. In order to evaluate the contributors' reputation, their asset value projections are automatically categorized by kind, and their likelihood of success is estimated by comparing the forecasts with real market data. [25]

For further improvement in the interpretability of of neural networks, In an effort to make the underlying neural networks easier to understand, An innovative FE-RNN architecture was presented. With a shared set of input and output language ideas, The structure is a parallel deep network composed of a POPFNN and an RNN. RNN employs fuzzy rules in the embedded POPFNN to associate the inference processes that are carried out. The interpretability of the hybrid network's inference process is improved through fuzzy IF-THEN rules. Employing RNNs within an FNN structure to model fuzzy entailment enables efficient data-driven implication. [13]

In energy markets, A way to use interpretable interval prediction to smoothly integrate the cross-energy and electricity markets was proposed. To begin with, a number of exogenous and multitemporal features from different significant energy markets are offered in an effort to clarify the aspect that is increasing the abnormal power costs. Next, in order to detect and predict the power price outlier sequence, a hybrid detector is presented that combines the Gaussian mixture model (GMM) with a lightweight gradient boosting machine. In addition, a hybrid LSTNet-kernel density estimate that is outlier-adaptive is suggested to offer interpretable interval prediction. In order to finally explain exogenous latent characteristics, the GMM-anticipated outliers strategy combines with the Shapley additive explanations method, with the assistance of the KDE in capturing uncertainty. LSTNet facilitates the integration of multitemporality across marketplaces and forecast the key trends [10]

Building upon the concept of interpretability, a novel hybrid model called ResTCN-DAM, which combines the advantages of dual attention mechanisms, temporal convolutional networks, and deep residual networks was proposed. The three modules' distinct characteristics are intended to be utilised by the suggested ResTCN-DAM: TCN is incredibly good at processing time series data in parallel. Multiple TCN layers can be densely layered to capture more hidden information in the temporal dimension by merging with a modified ResNet. With low computational overhead, the DAM module skillfully captures the interdependencies in both the temporal and feature dimensions, highlighting important time steps and characteristics while reducing less important ones. Moreover, the snapshot ensemble approach guarantees the precision and resilience of the forecasts by obtaining the result of training several models in a single training session. The deep integration and collaborative cooperation of these modules comprehensively enhance the model's forecasting capability from various perspectives. Visualization techniques are used to display heatmaps of the model's weights, thereby enhancing the interpretability of the model [26]

A novel interpretable sparse system identification approach was proposed that does not require any previous information. Rather of randomly using polynomial functions, as is the case with most system identification techniques, this method uses the Fourier transform to eliminate unnecessary entries from the dictionary matrix. It significantly lowers computational costs and displays a visual depiction of the system. A sparse description of the system model may be obtained by regularising the parameter matrix using the l_1 norm. [27]

3. Methodology

Algorithm for Comparative Analysis of Explainable Deep Learning Models using LIME and SHAP

Algorithm 1 Comparative Analysis of Explainable Deep Learning Models using LIME and SHAP

Require: Deep learning model M , dataset D , target variable y

Ensure: Comparative analysis report

```

1: Train the deep learning model  $M$  on dataset  $D$ 
2: Initialize LIME and SHAP explainers
3: Sample a subset  $S \subset D$  for explanation
4:  $LIME_{explanations} \leftarrow \{\}$ 
5:  $SHAP_{explanations} \leftarrow \{\}$ 
6: for each instance  $x \in S$  do
7:    $Z \leftarrow \{\}$  ▷ Perturbed samples
8:   for  $i = 1$  to  $k$  do
9:      $z_i \leftarrow Perturb(x)$  ▷ Generate perturbed sample
10:     $Z \leftarrow Z \cup \{z_i\}$ 
11:   end for
12:    $f(Z) \leftarrow M(Z)$  ▷ Get model predictions for perturbed samples
13:    $\pi_x(Z) \leftarrow Kernel(x, Z)$  ▷ Compute similarity weights
14:    $g \leftarrow TrainInterpretableModel(Z, f(Z), \pi_x(Z))$  ▷ Fit interpretable model
15:    $explanation \leftarrow g(x)$ 
16:   append  $explanation$  to  $LIME_{explanations}$ 
17:   Initialize  $\phi \leftarrow \{0\}$ 
18:   for each feature  $i \in N$  do
19:     for each subset  $S \subseteq N \setminus \{i\}$  do
20:       Compute  $v(S)$  as the prediction of  $M$  on subset  $S$ 
21:       Compute  $v(S \cup \{i\})$  as the prediction of  $M$  on subset  $S$  with feature  $i$ 
22:        $\phi_i \leftarrow \phi_i + \frac{|S|!(|N|-|S|-1)!}{|N|!} (v(S \cup \{i\}) - v(S))$ 
23:     end for
24:   end for
25:   append  $\phi$  to  $SHAP_{explanations}$ 
26: end for
27: Compute metrics for evaluation
28:  $LIME_{metrics} \leftarrow ComputeMetrics(LIME_{explanations}, S, y)$ 
29:  $SHAP_{metrics} \leftarrow ComputeMetrics(SHAP_{explanations}, S, y)$ 
30: Compare LIME and SHAP metrics
31:  $comparison \leftarrow CompareMetrics(LIME_{metrics}, SHAP_{metrics})$ 
32: Generate comparative analysis report
33:  $report \leftarrow GenerateReport(comparison)$ 
34: Return  $report$ 

```

3.1. Simulation environment and parameter list

The experiment was executed on a standard HP laptop powered by an Intel i5 processor within the Google Colab environment, without GPU acceleration. The simulation environment details include TensorFlow 2.12, Python 3.10, and a RAM capacity of 8GB. No specific hardware optimizations, such as GPU or TPU support, were utilized.

The following user-defined parameters were employed in the experiment:

- Learning Rate: 0.001
- Batch Size: 32
- Epochs: 100
- Optimizer: Adam
- Loss Function: Mean Absolute Error (MAE)
- Models: Conv1D, LSTM, NBEATS, LTCN
- Evaluation Metrics: MAE, RMSE, MAPE
- Data Window Size: 7 (number of previous days considered for prediction)

3.2. Dataset Description

The datasets utilized in this study was sourced from publicly available sources. This study consists of three datasets. The primary focus of this study was Adani power dataset consisting of 10 years of historical stock price. For dataset generalizability we also used 20 year historical stock price of TCS and 30 year historical stock price of reliance

3.3. Conv1d

The Conv1d model used for stock prediction is made up of three layers: dense, conv1d, and lamda. Convolution in one dimension, or Conv1D, is a method for processing sequence data in deep learning models. Conv1D takes a 1-dimensional array as input. This could represent time series data (e.g., sensor readings over time), a DNA sequence, or a text string. The core of Conv1D is a sliding window, called a kernel or filter, that moves across the input sequence. This filter contains learnable weights. The weights of the filter are multiplied element-wise by the corresponding items in the input sequence at each sliding window position. After that, the products are added together to get a single output value.. This process is repeated as the filter slides across the entire input sequence, generating a new 1-dimensional array called the feature map. The feature map captures the presence and strength of specific patterns detected by the filter within the input sequence. In practice, Conv1D uses multiple filters, each looking for different features. Each filter generates its own feature map, resulting in a higher-dimensional output that incorporates various aspects of the input sequence. Conv1D is powerful for extracting features from sequential data. It is most commonly used in tasks like Time series forecasting e.g., predicting stock prices, signal processing e.g., filtering noise from audio. By understanding how Conv1D works, it can be leveraged its capabilities for various sequence analysis problems.

3.4. LSTM

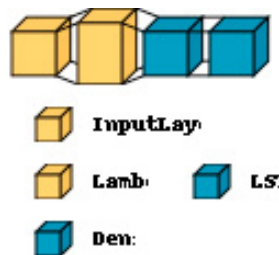


Fig. 1: LSTM

The LSTM model used for stock price prediction consists of input layer, lamda layer, dense layer and a lstm layer as shown in figure 2. LSTM is a specialized recurrent neural network adapt at capturing long-term patterns in sequential data. In LSTMs, the information flow within the memory cell is regulated by three crucial gates. They take into account both the input and the preceding output when deciding which data to exclude from the cell's current state.

decides what additional data should be added to the cell's state. determines which cell state data will be output as part of the concealed state for the subsequent time step. In simple terms, the forget gate decides what to forget, the input gate selects what new information to remember, and the output gate controls what remembered information to use. This gated mechanism allows LSTMs to learn and retain relevant information over long sequences Figure 2 displays the LSTM model's architecture. The input Layer takes a sequence of data. Lambda Layer Reshapes the input to (1,window size) to be compatible with LSTM layers. LSTM Layer A Long Short-Term Memory (LSTM) layer, equipped with 128 neurons and ReLU activation function, processes the time-series input data. The dense Layer is a fully connected layer with HORIZON units (number of prediction steps), mapping the LSTM output to the predicted values. Due to its proficiency in capturing long-term dependencies, LSTM is well-suited for predicting stock prices. It learns patterns in the historical data within the window size and uses this knowledge to predict future values (HORIZON steps ahead). ReLU activation is used in the LSTM layer to introduce non-linearity. Mean Absolute Error (MAE) loss and the Adam optimiser are used in the model's compilation for training. It's trained on 'train windows' and 'train labels' with validation on 'test windows' and 'test labels'.

4. Evaluation

This study leverages three distinct datasets, each comprising daily stock information for companies listed on the National Stock Exchange (NSE) of India. The datasets include ADANI Power, TCS, and Reliance Industries. The ADANI Power dataset contains 2,473 records with 13 features, while the TCS dataset consists of 4,898 records with 13 features. In contrast, the Reliance Industries dataset comprises 7,617 records with a single feature. These datasets provide a comprehensive basis for evaluating stock performance across different companies.

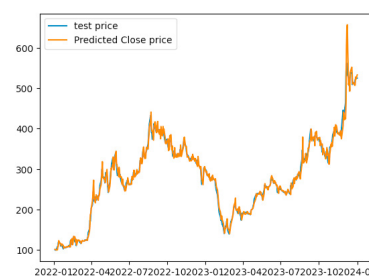
All the datasets were preprocessed. For all the datasets, The data preprocessing involved:

- Removal of unwanted columns
- Removal of Naan values
- Conversion into time series from multiple features to a single feature which is close price for price prediction
- Converting time series into a window size-based time series

The time series is has been made as time steps in which each one-day lag along with six days data is fed into the models to get the prediction output which means past 7 days data is fed into the neural network with with "feature 6" being price of the stock the day before and "Feature 0" being the price of the stock 7 days before then time series being split as 80 and 20 percent making for training and testing for feeding into nueral networks



(a) Conv1d Evaluation



(b) LSTM Evaluation

Fig. 2: Model Evaluation

The comparative analysis of Conv1D and LSTM models for stock price prediction is presented in Figures 2 and 3, highlighting their respective strengths and limitations. Both models demonstrate satisfactory predictive performance; however, quantitative evaluations from the comparative table reveal that Conv1D slightly outperforms LSTM for Adani Power. This marginal superiority of Conv1D can be attributed to the relatively short time series data and limited training set, where Conv1D's capability in capturing localized patterns provides an advantage. In contrast, LSTM, known for modeling long-term dependencies, shows improved performance on datasets with larger records

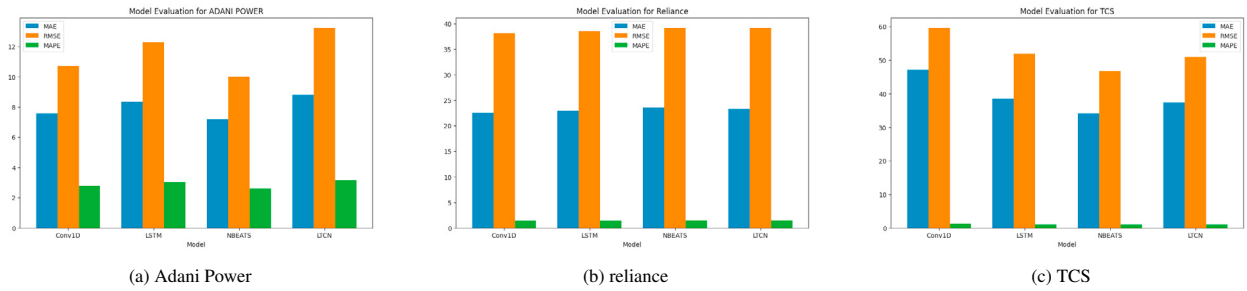


Fig. 3: Model Comparison

Evaluation	ADANI POWER			TCS			Reliance-industries		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Conv1D	7.58	10.71	2.8	47.2	59.67	1.36	22.48	38.08	1.40
LSTM	8.36	12.28	3.04	38.52	51.90	1.11	22.98	38.54	1.43
NBEATS	7.19	10.02	2.6	34.13	46.72	0.99	23.58	39.19	1.47
LTCN	8.8	13.24	3.14	37.38	50.89	1.11	23.27	39.14	1.46

Table 1: Evaluation table

such as TCS. Notably, From table 1 and bar plot of models NBEATS consistently delivers the best overall performance, achieving the lowest MAE, RMSE, and MAPE values, particularly for ADANI Power and TCS, indicating higher prediction accuracy and reduced errors. The LTCN model exhibits behavior similar to LSTM, likely due to architectural similarities. In the case of TCS, NBEATS demonstrates the most significant improvement, underscoring its strength in modeling complex stock price trends. However, for Reliance Industries, the performance gap between the models is minimal, with Conv1D and LSTM delivering comparable results, suggesting that either model could be suitable depending on specific constraints such as computational efficiency.

Cross-validation was performed, and the models were tested on an independent subset to assess their robustness across different data segments for Adani Power. The MAE scores across the five folds for Conv1D were 0.6367, 0.8333, 1.1277, 2.9283, and 7.2994, yielding an average MAE of 2.5651. These results indicate that Conv1D performs well on the first three folds but exhibits higher errors in the last two, particularly in the fifth fold, where the MAE surges to 7.2994. This suggests that Conv1D is sensitive to specific data characteristics, such as outliers or structural variations, which may hinder its performance on certain subsets. Similarly, the LSTM model's MAE scores across the five folds were 0.6251, 0.8368, 1.1212, 3.0950, and 10.0928, resulting in an average MAE of 3.1542. Like Conv1D, LSTM performs well on the initial folds but encounters difficulties in the fourth and fifth folds, with a pronounced error spike in the fifth fold (MAE of 10.0928), indicating that both models are susceptible to variations in the data across different segments.

4.1. SHAP

4.1.1. SHAP heatmap

Figure 7 and Figure 8 represents the shap heatmap for Conv1d and LSTM respectively. The SHAP (SHapley Additive exPlanations) heatmap provides insights into how each feature (time step in the input window) contributes to the model's predictions. A distinct occurrence from the test data for each characteristic is represented by each row in the heatmap. From feature 0 being 7 days before to feature 0 being the price of stock just before the day. Each cell's colour represents how that feature affects the model's prediction for that specific occurrence. Variables that drive the forecast higher that is, a higher projected price are shown by red cells, whereas variables that push the prediction lower that is, a lower projected price are indicated by blue cells. The color's intensity indicates how much of an impact there will be. By visually analysing the SHAP heatmap, it is clear that feature 6 (which is the just the day before) is the



Fig. 4: Heatmaps

most influential feature out of all the features for both the models and feature 2(4 days before) is the least influential in model prediction. These features 5 and 6 (one day before and 2 days) have a positive and negative impact while feature 2 (4 days before stock prediction) have no impact on model predictions.

4.1.2. SHAP Summary plot



Fig. 5: summary plot

Figure 9 and Figure 10 represent the shap summary plot for Conv1d and LSTM respectively. An overview of feature significance and how it affects the model's predictions is given by the SHAP summary plot. An individual prediction from the test data is represented by each dot on the plot. The SHAP value, shown by the x-axis, shows how the related feature affects the model's output. Negative SHAP values decrease the forecast, while positive ones increase it. The characteristics (time steps in the input window) are listed on the y-axis in order of overall relevance. The top row displays features 5 and 6, which are the most significant characteristics. The bottom row displays feature 2, which is the least significant feature for both model predictions. Each dot's colour corresponds to the related feature's value for that particular instance. Blue denotes lower feature values, whereas red denotes greater values.

4.1.3. SHAP Waterfall plot



Fig. 6: waterfall plot

Figure 11 and Figure 12 represents the waterfall plot shape for LSTM and Conv1d, respectively. The SHAP waterfall graphic provides a detailed description of how each feature (time step in the input window) influences the model's prediction for a specific occurrence, in this example, the first incidence in the test data. The base value of the model, or the average forecast made for every case, is shown on the left side of the plot. Every right-side bar indicates a single attribute's impact on the prediction. Each bar's length represents the contribution's magnitude, and its colour designates whether it is positive (red, pushing the prediction higher) or negative (blue, pushing the prediction lower), with feature 3 and feature 1 for conv1d and feature 1 for LSTM being the examples of positive contributions. Based on their entire contribution to the model, the features are arranged from top to bottom, with features 5 and 6 for both models being the most influential characteristics and feature 2 representing the least important. The instance's final prediction, which is the total of each feature contribution and the base value, is displayed on the right.

From shap heatmaps, shap summary plot and shap waterfall plot we can infer that shap feature 2 that is the feature that is least contributed to the models prediction whereas feature 6 has the highest significance on model prediction

We acknowledge the computational costs associated with applying SHAP to models such as Conv1D and LSTM, particularly in relation to the test data of shape (494, 7). Specifically, the execution time for SHAP was 748.03 seconds for the Conv1D model and 837.87 seconds for the LSTM model which signifies the fact that shap is computationally expensive

4.2. LIME

The LIME visualization plots of Conv1d and LSTM, respectively, are presented in figures 13 and 14, and The graphs illustrate the impact of individual features on the model's prediction for a specific event. The length of each bar corresponds to the feature's contribution to the overall forecast. Features that drive the forecast lower are indicated by blue bars, while those that push it higher are indicated by red bars. Each bar's text displays the related feature's real value in the example being described. The model's forecast for the instance is shown at the top of the plot. In figure 10 for Conv1d, it seems like features "Feature 6" have the most significant positive contributions to the prediction for both Conv1d and LSTM whereas, while "Feature 3" has the most significant negative contribution for Conv1d and "Feature 2" negative and least significant contribution for LSTM in Figure 11. Overall, the LIME plot helps in understand which features are driving the model's prediction for a given instance, providing local interpretability.



Fig. 7: LIME

5. Conclusion and future directions

From the conducted experiments and evaluations, it is evident that Nbeats outperformed all the other models in stock price prediction. The evaluation using XAI revealed that while recent data (the day prior) had the most significant impact on predictions, data from four days earlier had minimal influence on the model's forecast accuracy. However, a notable computational burden was observed when applying SHAP for explainability, even on small datasets, underscoring its computationally expensive nature. This finding suggests that in highly dynamic financial applications such as high-frequency trading, post hoc explainability approaches like SHAP may not be the most suitable tools due to their intensive computational demands.

The findings emphasize the need for further exploration into handling outliers and complex variations in stock price data. Future work could focus on integrating more robust preprocessing techniques or hybrid models that adapt to different data characteristics, enhancing the predictive stability of Conv1D and LSTM models across various data subsets. Additionally, exploring the application of NBEATS for multivariate time series analysis, particularly in dynamic financial scenarios like high-frequency trading, offers a promising direction for future research.

While precise stock price prediction remains elusive, explainable AI (XAI) holds considerable promise in enhancing trust and transparency in stock forecasting models. Techniques like SHAP and LIME not only generate predictions but also provide insights into the factors driving those predictions. This understanding empowers investors to make better-informed decisions by offering a clearer view of the reasoning behind expected price movements. Despite the challenges, explainable stock price forecasting has the potential to revolutionize financial research, fostering more confidence in AI-driven models.

As the field advances, XAI will likely play a critical role in the development of more reliable and transparent predictive models, contributing to a more informed and responsible investment landscape. Future research should focus on explainable multivariate time series analysis, as real-world financial data often involves multiple interacting variables rather than univariate time series. Additionally, exploring the interpretability of dynamic, high-frequency applications such as high-frequency trading offers valuable avenues for further investigation.

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